

A-053

Ty:- Prince. dzns.

Roll No. 038

Total No. of Pages.: 04

21N501 /

B.Tech. II Sem Main/Back (New Scheme) Acad. Session 2023-24

All Branch

(2FY1-01) – Engineering Mathematics-II

21N501/

Time : 3 Hours

Maximum Marks: 70

Min. Passing Marks:

Instructions to Candidates:

Part – A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All 10 questions are compulsory.

Part – B: Analytical/Problem Solving questions 5x4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving Design questions 3x10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1 Calculator

2 _____

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PART-A

- Q.1. Evaluate $I = \int_0^{\pi/2} \sin^5 x \cos^8 x \, dx$. [2]
- Q.2. Change the order of integration $\int_0^1 \int_x^1 e^{y^2} dy dx$. [2]
- Q.3. If $\vec{r} = xi + yj + zk$, then show that $\nabla \left(\frac{1}{r} \right) = -\frac{\vec{r}}{r^3}$. [2]
- Q.4. State the Gauss divergence theorem. [2]
- Q.5. The acceleration of a particle at any time $t > 0$ is given by $\vec{a} = 12 \cos 2t \hat{i} - 8 \sin 2t \hat{j} + 16t \hat{k}$ if velocity $\vec{v}$ and displacement $\vec{r}$ are at $t=0$, find $\vec{v}$ at any time. [2]
- Q.6. Show that the field defined by $\vec{a} = (x^2 + xy^2)\hat{i} + (y^2 + x^2y)\hat{j}$ is irrotational. [2]
- Q.7. Find the centre and radius of the sphere $2x^2 + 2y^2 + 2z^2 - 3x + 5y - 4z = 13$. [2]
- Q.8. Define Cone and Right Circular Cone. [2]
- Q.9. Find the rank of the matrix. [2]
- $$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$
- Q.10. Write the statement of Cayley-Hamilton's theorem. [2]

PART-B

- Q.1. Find the surface of the solid generated by the revolution of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ about the x-axis. [4]
- Q.2. Find the volume in the first octant bounded by the parabolic cylinders $z = g - x^2, x = 3 - y^2$ [4]
- Q.3. Evaluate the line integral $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$, [4]

Where C is the square formed by the lines $x = \pm 1, y = \pm 1$.

- Q.4. Using Green's theorem, evaluate $\oint_C x^2 y dx + x^2 dy$ [4]

Where C is the boundary (described counter clockwise) of the triangle with vertices (0, 0), (1, 0) and (1, 1).

- Q.5. Find the equation of the cone whose vertex is (3, 1, 2) and base is the circle $2x^2 + 2y^2 = 1, z = 1$. [4]

- Q.6. Reduce the matrix [4]

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

in its normal forms.

- Q.7. Investigate the value of λ and μ so that the equations [4]

$$2x + 3y + 5z = 9, \quad 7x + 3y - 2z = 8, \quad 2x + 3y + \lambda z = \mu$$

- have 1. No solution
2. Unique solution

PART- C

- Q.1. Evaluate the following integral by changing to polar co-ordinates: [10]

$$\int_0^1 \int_x^{\sqrt{2x-x^2}} (x^2 + y^2) dx dy.$$

- Q.2. Find the values of a, b and c such that $\vec{A} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational vector field. Also, find its scalar potential. [10]

- Q.3. Verify Stoke's theorem for $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ taken round the rectangle bounded by the lines $x = \pm a, y = 0, y = b$. [10]

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Q.4. Find the equation of the sphere which touches the plane $3x + 2y - z + 2 = 0$

at the point $(1, -2, 1)$ and cuts the sphere $x^2 + y^2 + z^2 - 4x + 6y + 4 = 0$ orthogonally. [10]

Q.5. Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

Hence find A^{-1} . [10]

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